「Topological Group.

「Def. Let G be a Group under the binary operation $\cdot : G\times G \to G$.
A Topological Space $(G, \mathcal{U})$ satisfying T_1 condition, and

$G\times G \to G : (x,y) \mapsto x\cdot y$ and $G \to G : x \mapsto x^{-1}$ are Continuous

is called the Topological Group.

「Lemma. Let G be a group and T_1-space. Then
G is Topological Group if and only if $G\times G \to G : (x,y) \mapsto xy^{-1}$ is continuous.

Proof. Suppose that G is Topological Group. Then
$\varphi(x,y) = xy$ and $\psi(y) = y^{-1}$ are continuous, therefore

$(x,y) \mapsto xy^{-1} = \varphi(x, \psi(y))$ is continuous, being composition of continuous maps

Conversely, suppose that $\mu : G\times G \to G : (x,y) \mapsto xy^{-1}$ is continuous.
Then, for fixed $x = 1$, $\mu(1,y) = y^{-1}$ is continuous, so $x \mapsto x^{-1}$ is continuous.
Now, $y \mapsto y^{-1}$ is continuous, so

$(x,y) \mapsto x\cdot(y^{-1})^{-1} = xy$ is continuous, being composition of continuous.

∎

「$GL_n(\mathbb{R})$ cf. SNU awl 69p

「$GL_n(\mathbb{R})$.

Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$, with Euclidean space.

Define a Metric for $M_{n\times n}(\mathbb{R})$ as;

$$d : M_{n\times n}(\mathbb{R}) \times M_{n\times n}(\mathbb{R}) \to \mathbb{R} ; (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu ; M_{n\times n}(\mathbb{R}) \times M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R}) : (A, B) \mapsto \left(\sum_{k=1}^{n} A_{ik} \cdot B_{kj} \right)_{ij}$$

is Continuous, because it is entry-wise Continuous.

Since $GL_n(\mathbb{R})$ is closed under the multiplication, it has also Continuous operation.

And,

$$GL_n(\mathbb{R}) \to GL_n(\mathbb{R}) : A \mapsto A^{-1}$$

is also Continuous, because $A^{-1} = (\det A)^{-1} \cdot \mathrm{adj}\, A$

where $(\mathrm{adj}\, A)_{ij} = (-1)^{i+j} \cdot \det A(j|i)$.

is also multivarlabes polynomial, entrywise.

⌈$SL_n(\mathbb{R})$

⌈Consider $O_n(\mathbb{R}) = \{A \in M_{n\times n}(\mathbb{R}) \mid A^t = A^{-1}\}$, $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$.
Then, O_n and SO_n are compact, and SO_n is connected.
proof. $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^{n} A_{ik} \cdot A_{jk} = \delta_{ij}$.

Define a map $f_{ij} : M_{n\times n}(\mathbb{R}) \to \mathbb{R} : A \mapsto \sum_{k=1}^{n} A_{ik} \cdot A_{jk}$.

Then, it is continuous therefore

$$O_n(\mathbb{R}) = \bigcap_{\substack{1\le i,j\le n \\ i\ne j}} f_{ij}^{-1}[\{0\}] \cap \bigcap_{1\le i=j\le n} f_{ij}^{-1}[\{1\}]$$ is finite intersection of closed set,

thus closed. Moreover, $\operatorname{diam} O_n(\mathbb{R}) \le n$, so it is bounded. Therefore, compact.
And, $\det : O_n(\mathbb{R}) \to \mathbb{R} : A \mapsto \det A$ is continuous, so,

$SO_n(\mathbb{R}) = \det^{-1}[O_n(\mathbb{R})]$ also closed, and bounded by subset of bounded set.

$\Gamma O_n(\mathbb{R})$

$\Gamma SOn(\mathbb{B})$

$\Gamma U_n(\mathbb{C})$